

The Heavens Scaled Down

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Abstract

The goal of this project is to build a scaled down Solar System with boundaries from the Earth and its moon, Luna, to the star of the Solar System, the Sun. Obviously, gravity will be a huge part of this system; the intention is to use Lagrangians instead of Newtonian mechanics, especially because there will be some references to general relativity. The codebase will be constructed from Ada for the system, Python for scripting, and a Bash environment. Documentation will be done with T_EX. This will be a cross-tool effort involving Claude AI, InsightsGPT, and Replit. To start, Chapter 2.2 and Appendix C of *Statistical Orbit Determination (2004)* by Byron Tapley, Bob Schutz, & George H. Born will be noted (many of the sentences will be directly quoted from the book). **Then, . . .**

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1 Introduction

2 The Fundamental Mechanics

This section explores the fundamental math and physics required for the proposed system. Particularly, §2.1 contains notes on Chapter 2.2 of [Tapley et al. \(2004\)](#), and §2.2 is a summary of its Appendix C [1].

2.1 General Properties of the Two-Body Problem

Although a point mass Earth is a serious stretch of the imagination, a spherical body with a constant, uniform mass distribution can be shown to be gravitationally equivalent to a point mass. Thus, as a more realistic approximation of the physical characteristics of a planet, a spherical body will be used.

Consider the motion of two spheres of mass M_1 and M_2 , as illustrated in Figure 2.1. No restrictions are placed on the magnitude of these masses; so, for illustrative purposes, assume M_1 represents the Earth and M_2 represents a satellite, such as Luna. The coordinate system is inertial and nonrotational. The equations of motion derived from Newton's laws of motion and law of gravitation are:

$$M_1 \ddot{\vec{R}}_1 = \frac{GM_1 M_2}{r^3} \vec{r} \quad (2.1a)$$

$$M_2 \ddot{\vec{R}}_2 = -\frac{GM_1 M_2}{r^3} \vec{r}. \quad (2.1b)$$

It follows that,

$$\ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r}, \quad (2.2)$$

where $\mu = G(M_1 + M_2)$ is the (*standard*) *gravitational parameter*. Even if M_2 were to be a satellite as large as Luna, then $M_{\text{Earth}} \sim M_{\text{Luna}} \times 10^{1.9}$; hence, $\mu \simeq GM_1$. Moreover, the general characteristics of the motion follow from manipulation of Equations (2.1).

The motion of a two-body system is defined by the position of the center of mass of the two spheres, $\vec{R}_{\text{cm}}$:

$$\vec{R}_{\text{cm}} = \frac{M_1 \vec{R}_1 + M_2 \vec{R}_2}{M_1 + M_2}. \quad (2.3)$$

Adding Equation (2.1a) and Equation (2.1b) gives:

$$\ddot{\vec{R}}_{\text{cm}} = 0 \quad (2.4)$$

$$\vec{R}_{\text{cm}} = \vec{C}_1(t - t_0) + \vec{C}_2, \quad (2.5)$$

where $\vec{C}_1$ and $\vec{C}_2$ are constants of motion. Equation (2.5) states that the center of mass of the two spheres moves in a straight line with constant velocity. Note that the straight line motion of $\vec{R}_{\text{cm}}$ is defined in an inertial coordinate system.

As for the angular momentum, it is obvious from Equation (2.2) that $\vec{r} \times \ddot{\vec{r}} = 0$; so,

$$\frac{d}{dt}(\vec{r} \times \dot{\vec{r}}) = 0. \quad (2.6)$$

Defining $\vec{h} \equiv \vec{r} \times \dot{\vec{r}}$, the *angular momentum per unit mass* (or *specific angular momentum*); $\vec{h}$ is called the angular momentum per unit mass because it is the angular momentum vector, $\vec{r} \times M_2 \dot{\vec{r}}$, divided by M_2 , which normalizes the angular momentum vector to a mass equal to one.¹ Anyway, from Equation (2.6), $\vec{h}$ is a **constant** with respect to time.

¹The angular momentum here is defined for the motion of M_2 (e.g., the satellite) with respect to M_1 (e.g., the Earth).

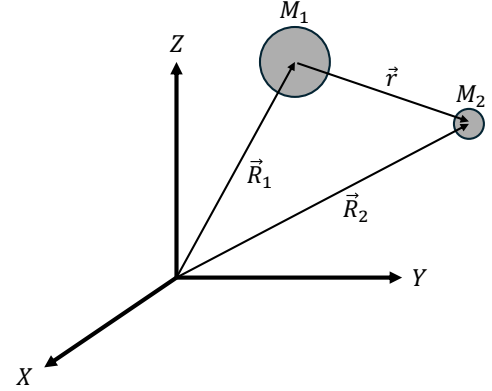


Figure 2.1: M_1 and M_2 are spheres of constant, uniform density. The position vectors of the spheres refer to their respective geometrical center, which also coincide with the center of mass of the sphere.

An important interpretation of this is that the orbital motion is planar, since the $\vec{h}$ vector must be perpendicular to the plane created by the $\vec{r}$ and $\dot{\vec{r}}$ vectors. The plane of motion is referred to as the *orbit plane*. Note that the specific angular momentum is defined for the motion of M_2 with respect to M_1 .

As for energy, consider the dot product of $\dot{\vec{r}}$ with Equation (2.2):

$$\dot{\vec{r}} \cdot \ddot{\vec{r}} = -\mu \dot{\vec{r}} \cdot \frac{\vec{r}}{r^3}, \quad (2.7)$$

by the chain rule, the LHS can be simplified to:

$$\dot{\vec{r}} \cdot \ddot{\vec{r}} = \frac{d}{dt} \left(\frac{\dot{\vec{r}} \cdot \dot{\vec{r}}}{2} \right). \quad (2.8)$$

On the other hand, note the following:

$$\frac{d}{dt}(r^{-1}) = \frac{dr^{-1}}{dr} \frac{dr}{dt} = -r^{-2} \frac{dr}{dt}, \quad (2.9)$$

then, recall that $r = \|\vec{r}\| = \sqrt{\vec{r} \cdot \vec{r}}$; so, calculating dr/dt ,

$$\frac{dr}{dt} = \frac{d}{dt} \left(\sqrt{\vec{r} \cdot \vec{r}} \right) \quad (2.10)$$

$$= \frac{1}{2\sqrt{\vec{r} \cdot \vec{r}}} (\dot{\vec{r}} \cdot \vec{r} + \vec{r} \cdot \dot{\vec{r}}) \quad (2.11)$$

$$= \frac{1}{2\sqrt{\vec{r} \cdot \vec{r}}} (2\vec{r} \cdot \dot{\vec{r}}) \quad (2.12)$$

$$= \frac{\vec{r} \cdot \dot{\vec{r}}}{r}. \quad (2.13)$$

Putting (2.13) into Equation (2.9) gives:

$$\frac{d}{dt} \left(\frac{1}{r} \right) = -\frac{\vec{r} \cdot \dot{\vec{r}}}{r^3}. \quad (2.14)$$

Now, Equation (2.7) can be written as:

$$\dot{\vec{r}} \cdot \ddot{\vec{r}} = \mu \frac{d}{dt} \left(\frac{1}{r} \right), \quad (2.15)$$

Equating the RHS of Equation (2.15) with the RHS of Equation (2.8) and integrating with respect to t leads to:

$$\xi = \frac{\dot{\vec{r}} \cdot \dot{\vec{r}}}{2} - \frac{\mu}{r}, \quad (2.16)$$

where ξ is mathematically a **constant** of integration, and physically, it is the scalar *energy per unit mass* (also known as *specific orbital energy*), implying that the total energy is conserved. Please note that ξ has the form of a Hamiltonian;

$$M_2 \xi \simeq \frac{1}{2} M_2 \dot{\vec{r}}^2 - \frac{GM_1 M_2}{r}, \quad (2.17)$$

where the expression for the potential energy is negative. Nonetheless, in total, there are ten constants of motion pertaining to the problem of two bodies, three vectors and one scalar: $\vec{C}_1$, $\vec{C}_2$, $\vec{h}$, and ξ .

2.1.1 Motion in a Plane

It has been shown that the orbital motion of M_2 with respect to M_1 takes place in a plane perpendicular to the angular momentum vector. Please refer to Figure 2.2 for an illustration of the plane of motion relative to polar and rectangular coordinates. As mentioned before, (X, Y, Z) is a nonrotating frame. The $\tilde{X}$ -axis is at the intersection of the orbit plane and the (X, Y) plane, the $\tilde{Y}$ -axis is perpendicular to the $\tilde{X}$ -axis but lies within the orbit plane, the $\tilde{Z}$ -axis is coincident with the specific angular momentum.

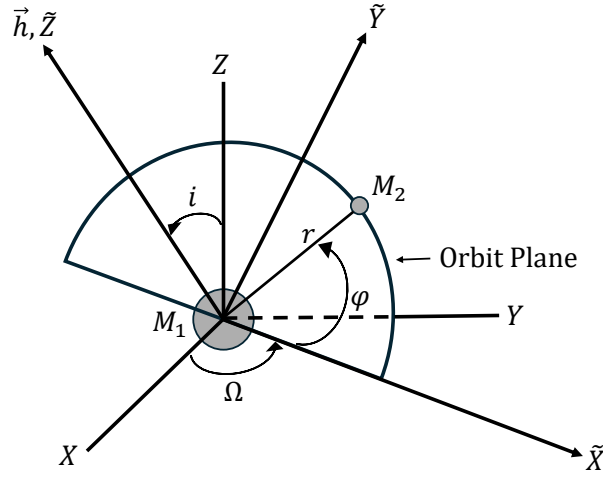


Figure 2.2: The orbit plane in space. Since the angular momentum is constant, the motion of M_2 with respect to M_1 takes place in a plane, known as the orbit plane. This plane is perpendicular to the specific angular momentum vector, $\vec{h}$, and the orientation is defined by the angles i and Ω .

The fact of planar motion can be used to find the solution to the equation of motion given from Equation (2.2). In polar coordinates, the velocity and acceleration are [2]:

$$\vec{r} = \dot{r}\hat{r} + r\dot{\phi}\hat{\phi} \quad (2.18)$$

$$\ddot{\vec{r}} = (\ddot{r} - r\dot{\phi}^2)\hat{r} + (r\ddot{\phi} + 2\dot{r}\dot{\phi})\hat{\phi}. \quad (2.19)$$

As seen in Figure 2.3, since $\vec{r} = r\hat{r}$, Equation (2.2) can be written in polar coordinates;

$$\ddot{\vec{r}} = -\frac{\mu}{r^2}\hat{r}, \quad (2.20)$$

matching the components of (2.19) and (2.20):

$$\ddot{r} - r\dot{\phi}^2 = -\frac{\mu}{r^2} \quad (2.21a)$$

$$r\ddot{\phi} + 2\dot{r}\dot{\phi} = 0. \quad (2.21b)$$

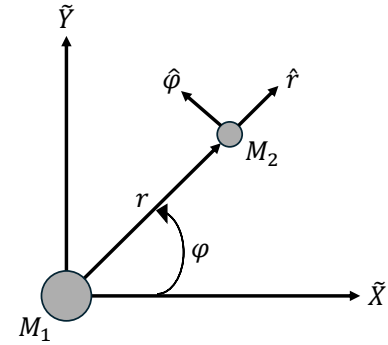


Figure 2.3: The planar motion described with polar coordinates.

These equations describe radial acceleration modified by centripetal acceleration and the conservation of angular momentum in the transverse direction, respectively. The transverse equation, Equation (2.21b), effectively shows that the change in angular momentum is zero, consistent with planar motion and constant angular momentum. Thus, Equation (2.2), expressed in polar coordinates, breaks into the scalar Equations (2.21), representing the radial and angular dynamics of the orbiting body.

When it comes to the velocity of the azimuthal angle, $\dot{\phi}$, there is a binary case of it being either zero or not. If $\dot{\phi} = 0$, then motion is rectilinear, which is a special and uninteresting case. If $\dot{\phi} \neq 0$, then orbital motion occurs.

Equation (2.21b) can be written as,

$$\frac{d}{dt}(r^2\dot{\phi}) = 0. \quad (2.22)$$

Following from the expression for the specific angular momentum and Equation (2.18):

$$\vec{h} = \vec{r} \times \dot{\vec{r}} \quad (2.23)$$

$$= (r\hat{r}) \times (\dot{r}\hat{r} + r\dot{\phi}\hat{\phi}), \quad (2.24)$$

the cross product $\hat{r} \times \hat{r} = 0$, and the cross product $\hat{r} \times \hat{\phi} = \hat{h}$ as $\hat{h}$ is perpendicular to the plane of motion; so,

$$\vec{h} = r^2\dot{\phi}\hat{h}. \quad (2.25)$$

From Equation (2.22), $h = r^2\dot{\phi}$ is a **constant**. Furthermore, in orbital mechanics, the true anomaly, ϕ (the angle describing the position of the orbiting body along its orbit), can serve as a natural independent parameter to describe the orbit's shape and position, especially when the specific angular momentum is constant. Consider the following substitutions in both the dependent and independent variables in Equation (2.21a):

$$r \longrightarrow \frac{1}{u} \quad (2.26a)$$

$$\phi \longrightarrow t. \quad (2.26b)$$

Originally, $r = r(t)$ and $\phi = \phi(t)$, so both radius and angle depend on time. However, because $\vec{h}$ is conserved and nonzero, $\phi(t)$ is a monotonic function of time t , which means that the azimuthal angle can serve as a new independent variable instead of time. The derivatives with respect to time translate into derivatives with respect to ϕ . This change simplifies the formulation of the two-body problem because it leverages angular momentum conservation and eliminates time explicitly, focusing on the geometric properties of the orbit. In any case, from Equation (2.25) and Equation (2.18), the speed in polar coordinates can be written as:²

$$v^2 = \dot{r}^2 + \frac{h^2}{r^2}, \quad (2.27)$$

Using the substitution (2.26a) and Equation (2.16) for the specific orbital energy, after some manipulation, the result is the second-order differential equation:

$$\frac{d^2u}{d\phi^2} + u = \frac{\mu}{h^2}, \quad (2.28)$$

which is known as the orbit equation for the two-body Kepler problem [3]. This equation describes how the radius changes as the body sweeps around an angle ϕ , leading directly to conic sections (ellipse, parabola, and hyperbola) as solutions. The expression for the solution of Equation (2.28) is:

$$u = \frac{\mu}{h^2} [1 + e \cos(\theta - \psi)], \quad (2.29)$$

where ψ is a constant of integration and e is the eccentricity as a function of energy and angular momentum;

$$e = \sqrt{1 + \frac{2\xi h^2}{\mu^2}}, \quad (2.30)$$

rewriting Equation (2.29),

$$\frac{1}{r} = \frac{\mu}{h^2} + A \cos(\phi - \psi), \quad (2.31)$$

²The dot product of any two perpendicular unit vectors is always zero.

where the constant A is:

$$A = \frac{\mu e}{h^2}. \quad (2.32)$$

From Equation (2.31), it can be readily shown that:

$$\frac{dr}{d\varphi} = \frac{\mu e \sin(\varphi - \psi)}{h^2 [1 + \cos(\varphi - \psi)]^2}. \quad (2.33)$$

When $\varphi = \psi$, or when $\varphi - \psi = \pi$, then $dr/d\varphi = 0$. The second derivative at $\varphi = \psi$ is always greater than zero;

$$\frac{d^2r}{d\varphi^2} > 0, \quad (2.34)$$

and so it is evident that at $\varphi = \psi$, r is a minimum.³ This location with minimum r is referred to as the *perifocus*, and only in the case of a satellite of the Earth, it is referred to as the *perigee*. As a consequence, ψ is known as the *argument of perigee*. Over and above that, introducing the *true anomaly*, $f = \varphi - \psi$. The expression for r can then be written as:

$$r = \frac{p}{1 + e \cos f}, \quad (2.35)$$

where $p \equiv h^2/\mu$.

³When the second derivative of a function is positive at a point, mathematically, it means the function is concave upward at that point (a shape like a cup). This concavity indicates that the point is a local minimum because the function decreases before the point and increases after it.

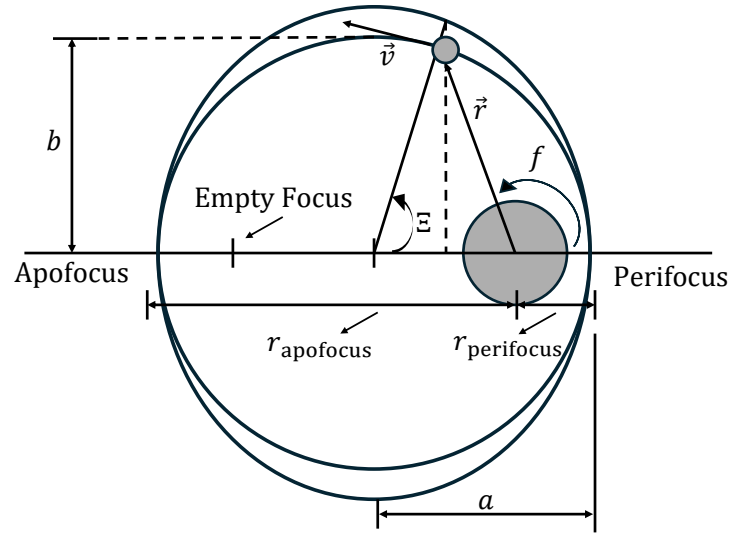


Figure 2.4: As before, M_1 is the central mass and M_2 is the satellite. The inner circle that M_2 is placed on in this illustration is not the actual path of the orbit, it is an auxiliary geometric construct called the auxiliary (or circumscribing) circle used to define the eccentric anomaly. The true orbit of M_2 is the ellipse, and $\vec{r}$ always points from M_1 to the true position of M_2 on the ellipse.

2.2 Lagrangian Equations of Motion

References

- [1] Tapley B. D., Schutz B. E., & Born G. H. (2004). Statistical Orbit Determination, *Elsevier Academic Press*. [ISBN-13: [978-0-12-683630-1](#), ISBN-10: [0-12-683630-2](#)]
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- [3] Wikipedia contributors. Kepler Problem, *Wikipedia, The Free Encyclopedia*, retrieved on 06/02/2026 from https://en.wikipedia.org/wiki/Kepler_problem#Solution_of_the_Kepler_problem.